

Fuzzy-Based Crime Prevention Strategies for Urban Safety

Final presentation for the Seminar on Urban Computing

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INITIAL IDEA

Our Approach

- Fuzzy-based Algorithms to process ambiguous data, allowing for **flexible crime prediction** and **dynamic resource management**.
- Engage the community, providing both law enforcement and citizens a voice in shaping safety strategies.
- Emphasizes social equity by avoiding biased predictions and over-policing.

Artifact Features

- **Risk Assessment:** Fuzzy logic can analyze various risk factors (e.g., previous crime rates, time & seasonality and event schedules) to identify areas at higher risk for crime. This allows for targeted interventions.
- **Predictive Policing:** Better predicting potential crime hotspots based on incomplete or uncertain data.
- **Community Engagement:** Taking qualitative feedback from community surveys, allowing police to understand public concerns more effectively and adjust strategies accordingly.
- **Resource Optimization:** Optimize general practices and schedules by the police based on various parameters like time of day, local events, and historical data, ensuring a more efficient use of police resources.

Artifact Goal

Create an easy-to-use platform that law enforcement, urban planners, and communities can utilize to make better decisions in crime prevention.

WHERE DOES THE ISSUE LIE

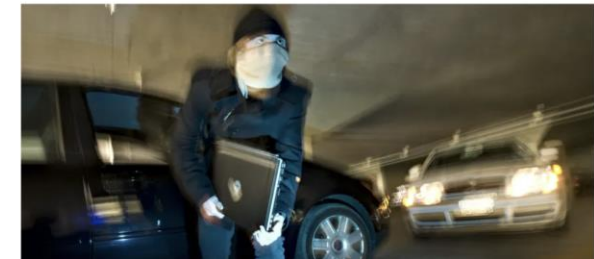
ANOTHER +16% BREAK-INS IN 2023 IN SWITZERLAND

Tuesday, 30.04.2024



News >
Life & Aging >

Switzerland sees record levels of theft in 2023



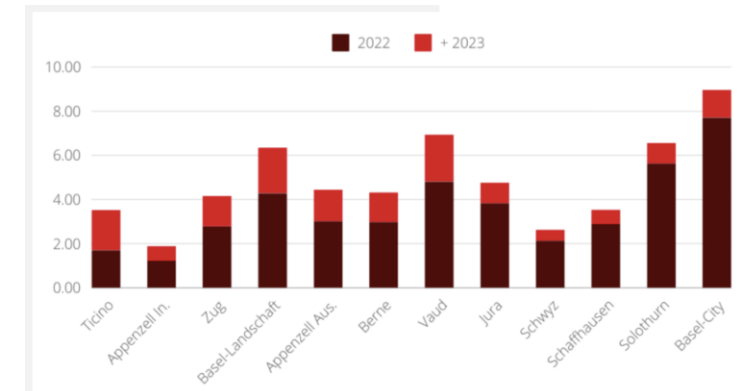
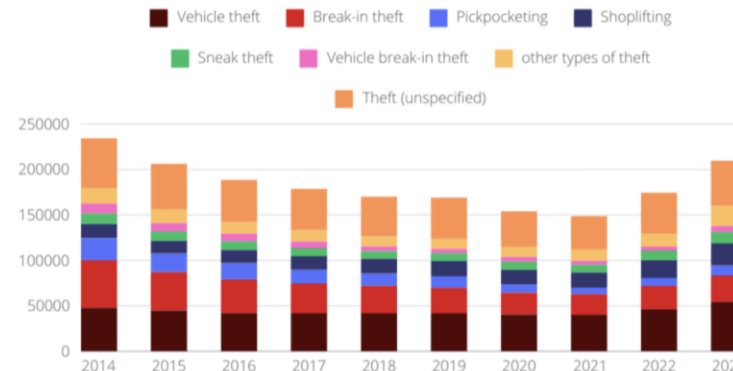
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- A total of 522 558 offences recorded under the Swiss Criminal Code (SCC).
- Increase was mainly **due to property offences (+17.6%)**, which rose for the second year in a row.
- **In 2023, the police recorded a daily average of 114 break-ins or sneak-in thefts**, compared to 96 such cases per day the previous year. This figure not only reflects the daunting challenges confronting security services, it also underscores the urgent need for proactive measures.

Story 1: One of my friends phone was stolen at the Bern Bahnhof. She asked people around if they have seen a phone. Obviously, she didn't find much help. She went to the police and got to know that it is holiday time and she will have to report the complaint 3 days from today.

Story 2: Another friend who faced theft in front of the university. His gold chain was snatched and also he faced some abuse.

Story 3: A friend's parcel went missing and when contacted with the parcel company, he suspects that it was stolen from the letter box area. Another event that happened at festive time. Statistics suggest that parcel theft is high in festive season.

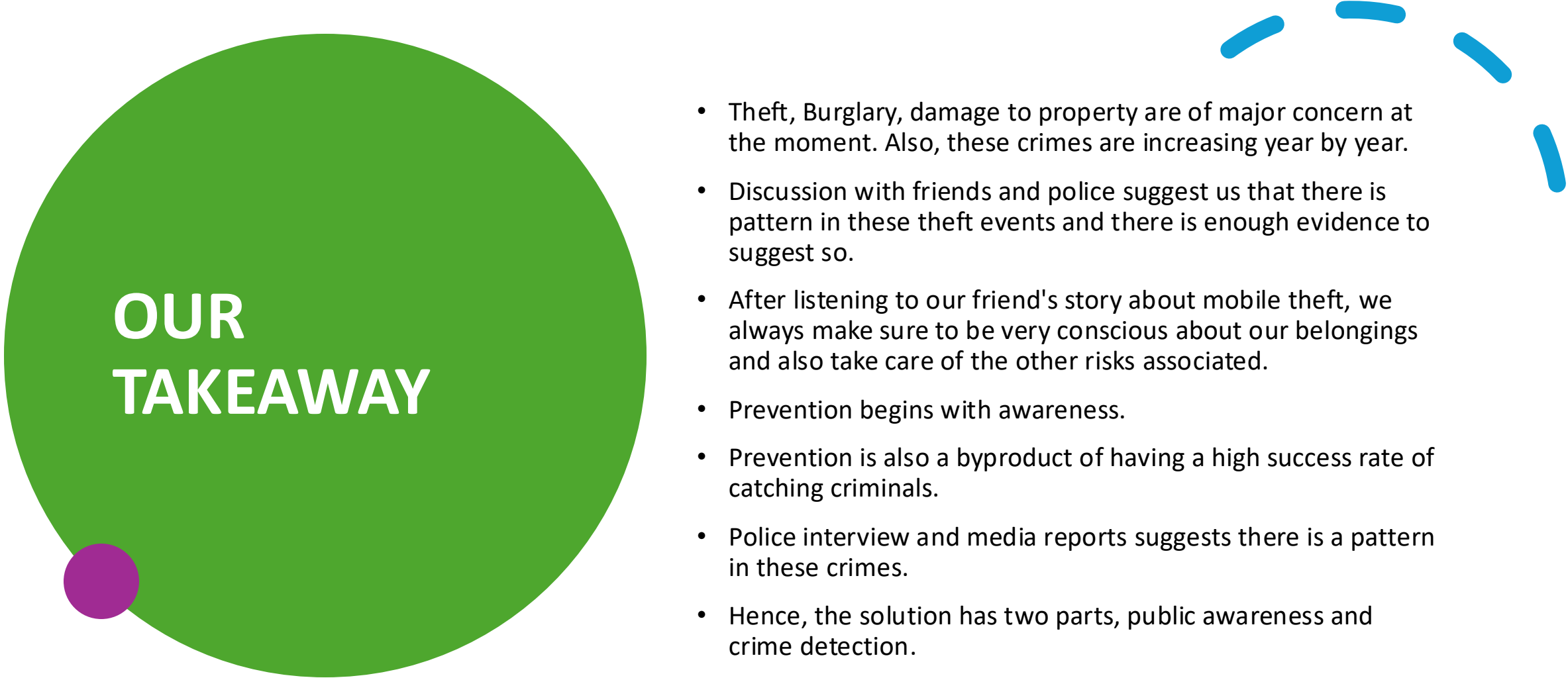
Story 4: Today, something really scary happened in Graubünden, a town that's super famous for skiing and winter holidays. Two hotels got robbed last night, and it's freaking people out. There's this person who has an office nearby with big windows that show everything inside, and they're really worried it might be targeted next. A while ago, they also heard about some really expensive stuff being stolen from a construction site for a railway tunnel. It's starting to feel like nothing is safe around here anymore!

EXPERIENCE FROM FRIENDS & COLLEAGUES



DISCUSSION WITH THE POLICE DEPARTMENT

- Q: Maximum Resource Consumption?
A: Reporting and Investigation.
- Q: Most common reports by people?
A: Theft and Burglary.
- Q: A lot of theft reports and news around important places like Bern Bahnhof, is that true?
A: Yes, crowded places end up with more theft reports.
- Q: Are officers overworked?
A: Not so much.
- Q: Does always the nearest station to the crime figure out the case itself?
A: Usually yes, but sometimes help from other stations is needed.
- Q: Do you think public awareness creates results?
A: Surely, precautions starts from an individual.
- Q: Is it common that thefts occur more in specific areas?
A: Yes, some factors are population of the area, socio-economic factors,
- Q: How aware are people about neighborhood tragedies?
A: Depends on person to person.
- Q: We are prototyping a hotspot detection application which can analyze patterns from past data such as previous crime rates, time of crime, population of an area, citizen traffic, major citizen traffic etc. And trying to predict severity at a particular location. Do you think this is beneficial?
A: Surely, we see patterns in these reports and as statistics suggest that theft is increasing, any measures that can help us reduce that, are going to be helpful.

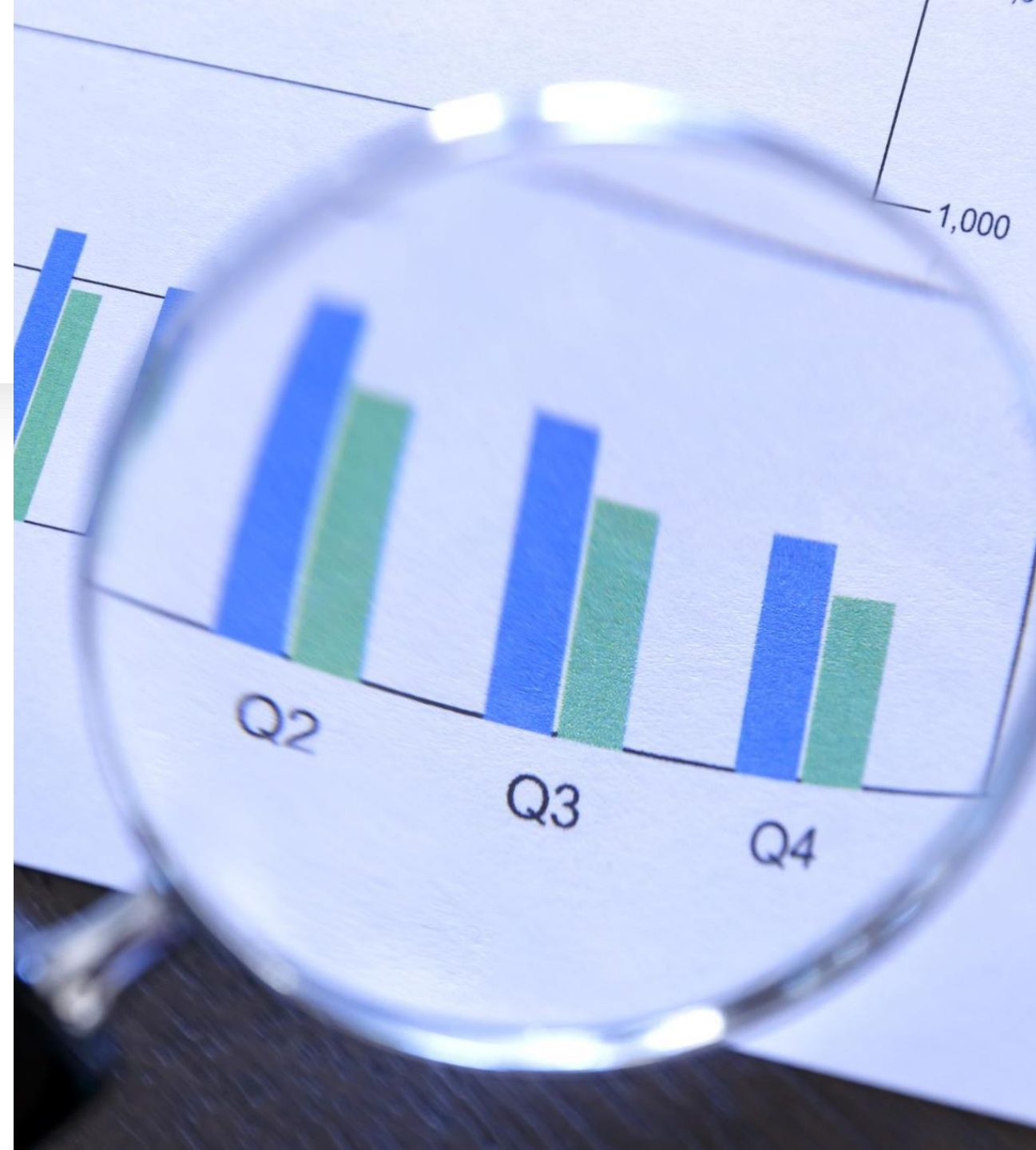


OUR TAKEAWAY

- Theft, Burglary, damage to property are of major concern at the moment. Also, these crimes are increasing year by year.
- Discussion with friends and police suggest us that there is pattern in these theft events and there is enough evidence to suggest so.
- After listening to our friend's story about mobile theft, we always make sure to be very conscious about our belongings and also take care of the other risks associated.
- Prevention begins with awareness.
- Prevention is also a byproduct of having a high success rate of catching criminals.
- Police interview and media reports suggests there is a pattern in these crimes.
- Hence, the solution has two parts, public awareness and crime detection.

DATA IN REPORTS

- **Date and Time**
- **Location**
- Name, age, gender, and contact details of the person filing the report.
- Identification details like ID card numbers are recorded.
- Detailed description, value and documentation of stolen items
- **Narrative:** Timeframe, Circumstances, Suspicious Activity around
- **Description of Suspects, Mode of Operation**
- **Evidence Collected**



SOLUTION DISCUSSION

- Platform 1: Public Awareness
 - We apply an automation procedure to share recent incident reports to the users who are susceptible to similar issues.
 - We do not share private information, but only what makes our citizens more aware.
 - We filter reports according to the threat levels so that the citizen does not get feared a lot by these reports. Ex, Elders should not be shared extreme crimes.
 - We act like a media who states facts and works for the favor of citizens.
- Platform 2: Hotspot Detection
 - Using the information from all the previous reports, we build a Neural Network model which can tell us risk levels of particular areas at specified time of day and time of the year.
 - The input features consists of numerical columns such as time, date, theft damage amount, age of victim, population of the neighborhood, foot count of any event taking place.
 - Information from categorical columns such as gender, location, mode of operation, risk level of the crime.
 - We build features from the text description provided by the victims.

FEATURES OF THE PLATFORM

Note: We cannot process any online data from the police government due to CORS restrictions. We built this prototype just to showcase our idea. For development purposes we will require their permission.

Purpose 1: Citizen Notification System

- 1. Dynamic Filtering:** Control over which information is sent and to whom. For example, Severe cases excluded from elderly.
- 2. Geographical Relevance:** Notify only the citizens in proximity to the theft location. Use geo-fencing to define the “area of interest.”
- 3. Frequency Management:** Limit notifications to avoid overburdening users. For example, notify only for significant thefts with a certain threshold of severity.

Technical Implementation

- 1. Database Setup:** Store reports with attributes - severity, location, victim demographics, etc.
- 2. Notification Engine:** A system that uses predefined rules or machine learning to filter and decide notification content. Include user preferences for notification frequency and type.
- 3. Delivery System:** Use mobile app push notifications or SMS alerts. Ensure GDPR compliance for data privacy.

User Segmentation:

- **Elderly Citizens:** Minimal notifications; exclude severe cases.
- **Young Adults:** More frequent, relevant updates.
- **Local Businesses:** Theft trends in the vicinity.

Purpose 2: Risk Prediction Using Neural Networks

Numerical Columns: Time, date, theft damage amount, age of the victim.
Population of the neighborhood, foot count of local events.

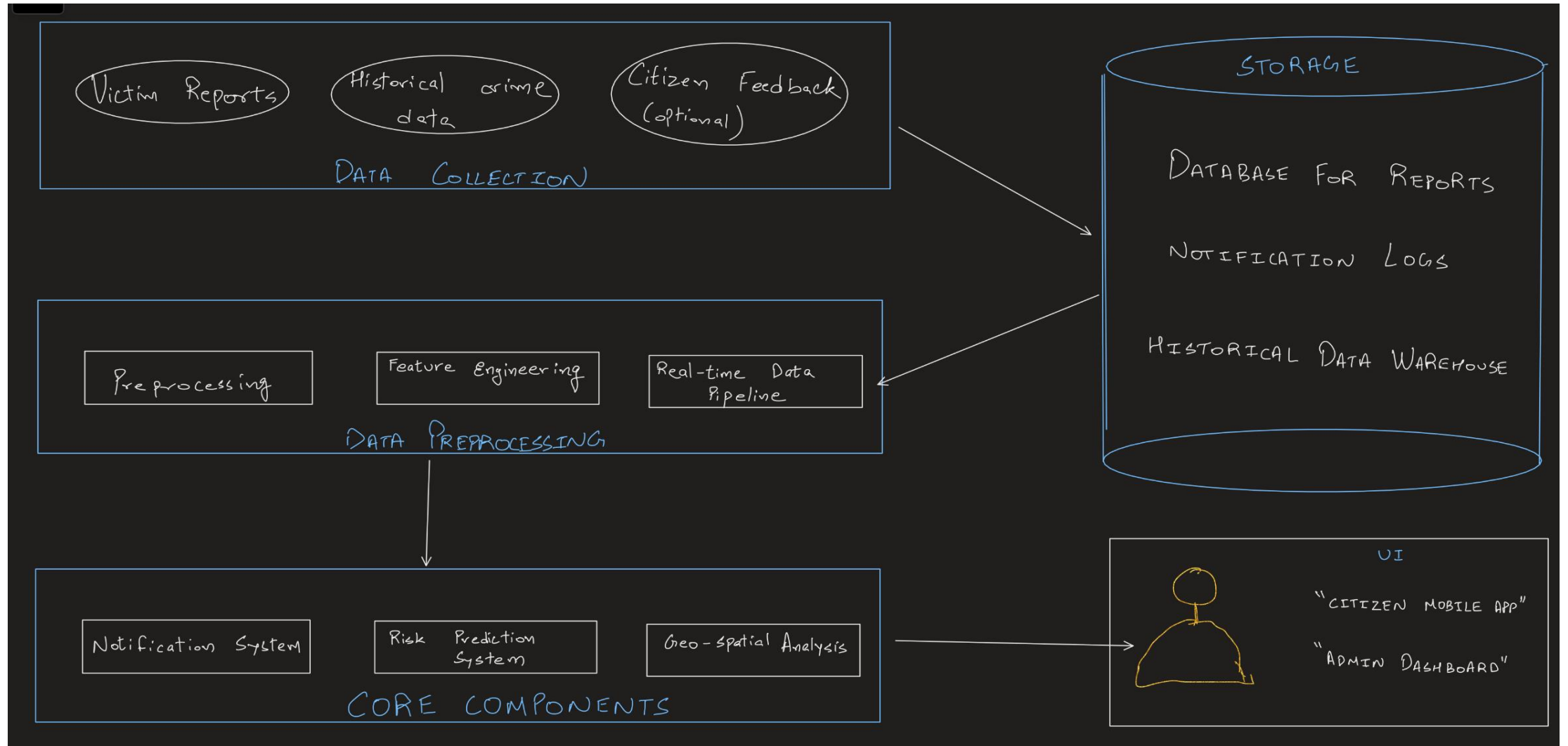
Categorical Columns: Gender, location, mode of operation, risk level of the crime.

Textual Data: Extract features using **NLP techniques** (e.g., TF-IDF, Word2Vec) from victim descriptions.

Model Design

- 1. Preprocessing:** Normalize numerical data. One-hot encode categorical columns. Use text embedding for victim descriptions.
- 2. Architecture:**
 - **Input Layer:** Numerical data directly. Embedded categorical and textual data.
 - **Hidden Layers:** Dense layers with activation functions like ReLU. Dropout layers to prevent overfitting.
 - **Output Layer:** Single neuron with a sigmoid/softmax activation for risk level prediction.
- 3. Training:** Train using historical theft data with timestamps and locations. Optimize with loss functions like cross-entropy for classification tasks.
- 4. Outputs:** Risk score for specific areas at a given time.

ARCHITECTURE



RESULTS OF SUCH SYSTEMS AT A SIGHT

1. United States (Citizen App):

- Increased public awareness and enabled citizens to avoid high-risk areas, but occasionally caused unnecessary panic due to unverified notifications. There were some insights from CCTV suggesting that people who received notifications took safety measures.

2. Netherlands (Predictive Policing - CAS):

- Led to a measurable reduction in burglaries and property crimes, but faced criticism for potential privacy invasion and algorithmic bias.

3. Denmark (POL-INTEL Project):

- Strengthened police collaboration and preempted organized crime activities effectively, though it relies heavily on accurate data and advanced infrastructure.

4. United Kingdom (AI in Prisons):

- Improved early detection of violent behavior and self-harm incidents, contributing to a safer prison environment, albeit with ongoing ethical concerns.

Real-time crime notification systems and predictive policing have yielded mixed outcomes. They effectively enhance safety and resource allocation but face challenges like ethical dilemmas, potential biases, and public trust issues. Balancing technology and human oversight is crucial for their long-term success.



CONCLUSION

This application aims to enhance public safety and reduce theft crimes by providing real-time crime notifications and predictive analytics. Utilizing victim reports and advanced neural networks, it predicts neighborhood risk levels to empower citizens with timely information and help authorities allocate resources effectively. While promising, challenges such as data privacy, algorithmic bias, and maintaining ethical transparency must be addressed to ensure responsible implementation. Lessons from similar global initiatives emphasize the importance of accuracy, diverse data handling, and citizen engagement without inciting panic. By balancing innovation with ethical considerations, this system aspires to foster sustainable urban safety.



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ANY QUESTIONS? 🤪

